Where the Model Wanders and the Mind Converges: A Distributional Comparison of Human and VARC Failures on ARC

Jing Yu Huijing Cong Wanyi Zhou

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Abstract

The Abstraction and Reasoning Corpus (ARC) remains one of the few benchmarks on which strong neural systems still trail human performance. Most prior analyses report task-level accuracy, treating each task as a binary pass/fail event. We argue that this hides a more fundamental difference in *how* the two agents fail. Using the official VARC ViT model's 510 multi-view predictions per task and 946 participants from H-ARC, we treat each side's predictions as a sample from an answer distribution and compare their entropy, top-1 vote share, and the number of distinct candidate grids. On the 111 tasks where humans succeed but VARC fails, humans converge on ≈ 3.6 distinct answers (top-1 share 74%, entropy 0.80); on the same tasks VARC scatters across ≈ 264 distinct grids (top-1 share 13%, entropy 4.72). The correlation between human and VARC entropy is only 0.21. We argue this signature reflects a *failure-to-induce*: rather than picking among a small hypothesis set as humans do, the model produces hundreds of pixel-space candidates, suggesting it never localizes a rule.

1 Introduction

The Abstraction and Reasoning Corpus [1] was designed as a deliberate counterpoint to large-scale supervised benchmarks: each task supplies only a handful of input–output examples and asks the solver to induce a transformation rule that generalizes to held-out inputs. Modern systems, including the VARC family of vision transformers [2], have closed much of the gap to human performance, but a substantial residual remains. The natural next question is not how much remains, but *what kind of failure accounts for it*.

Prior work largely reports per-task accuracy or pass@k scoring on ARC, which collapses each task to a single bit and obscures the underlying behavior. The H-ARC dataset [3] provides the complementary human-side signal: multiple participants attempting each task, with intermediate and final grids logged. Concurrently, VARC's inference procedure already provides an analogous signal on the model side: each test input is rendered through 510 augmented *views*, and the final prediction is a majority vote over the 510 view-level outputs. The 510-vote distribution is not normally examined directly, but it is exactly the distribution we would want if we were asking what hypothesis the model entertains.

We exploit this symmetry: ~ 10 human responses per task and 510 view predictions per task can both be summarized as a distribution over answer grids. We ask whether the *shape* of that distribution distinguishes the two agents on the tasks that matter most — those where humans succeed and the model fails.

Contributions.

- We introduce a passive input-intervention analysis that reuses VARC's existing 510-view predictions to characterize model confidence at task level, with no additional inference.
- We define matched answer-distribution metrics for human and model and compare them on H-ARC's 400 evaluation tasks.
- We identify a qualitative asymmetry: on tasks humans solve and VARC misses, humans concentrate on a handful of candidate grids while the model scatters across hundreds. We argue this is a *failure-to-induce* signature, not a fine-grained error.

2 Methods

2.1 Data

We use the 400-task ARC-AGI evaluation set as the shared benchmark. For the model side we use VARC's official ARC-1_ViT predictions, specifically the `attempt_0` folder, which contains 510 view-level predicted grids per test input. For the human side we use H-ARC's `summary_data.csv`, which contains 7820 attempt-level submissions from 946 participants. We restrict to `task_type == evaluation` throughout and take each participant's *last* attempt as their final answer for that task. Table 1 summarizes the data.

Source	**Count**
ARC evaluation tasks	400
VARC views per task (`attempt_0`)	510
H-ARC participants	946
H-ARC final answers (last attempt $\times$ task)	≈ 10 per task

Table 1: Data sources used in this study.

2.2 Correctness scoring

For VARC we follow the official plain-ViT scoring: within `attempt_0`'s 510 views per test input, count grid-level votes and retain the top-2 most-frequent grids; the task is correct if either of those two matches the ground truth (pass@2). For human correctness on a task we report `human_accuracy`, the fraction of participants whose final answer matches ground truth. To form the four quadrants in Section 3, we declare the human side "correct" on a task when `human_accuracy` > 0.5.

2.3 Error typing

For diagnostic purposes (independent of the pass@2 correctness calculation) we classify each predicted grid against ground truth into one of four categories:

- `correct`: exact match.
- `wrong_size`: row or column count differs from ground truth.
- `close_miss`: shape matches and cell-level accuracy ≥ 0.8.

- `wrong_content`: shape matches but cell-level accuracy < 0.8.

For VARC we use the top-1 voted grid for error typing, mirroring how the single most-likely model output would be inspected.

2.4 Answer-distribution metrics

The core methodological move is to treat the 510 view-level VARC predictions and the ~ 10 human final answers as samples from parallel *answer distributions* on each task. For each task and each agent we compute, over the empirical distribution of distinct grids $\{g_i\}$ with vote counts n_i (total N):

$$\begin{aligned}
\text{top-1 share} &= \frac{\max_i n_i}{N}, \\
\text{top-1/top-2 gap} &= \frac{n_{(1)} - n_{(2)}}{N}, \\
H &= -\sum_i \frac{n_i}{N} \log \frac{n_i}{N} \quad \text{(Shannon entropy, nats)}, \\
n_{\text{unique}} &= |\{g_i\}|.
\end{aligned}$$

For the model these summarize the 510-view output distribution; for humans they summarize the per-task distribution of last-attempt grids. Because both sides are scored on the same units (distinct grids, proportions of agents), the metrics are directly comparable.

2.5 Quadrant taxonomy

We partition the 400 tasks by joint correctness:

- `both_correct` ($n = 173$),
- `human_only` ($n = 111$): human majority correct, VARC pass@2 wrong,
- `varc_only` ($n = 48$): VARC correct, human majority wrong,
- `both_wrong` ($n = 68$).

The `human_only` group is the focus of our analysis: these are the tasks on which the model's failure is least excusable, since the problem demonstrably admits a solution that ordinary participants find.

3 Results

3.1 Aggregate accuracy and error type

Across the 400 evaluation tasks, mean human last-attempt accuracy is 64.1% and VARC pass@2 accuracy is 55.3%, matching the 55.63% plain-ViT pass@2 score reported by Hu et al. [2] on the ARC-1 evaluation set. The four quadrants from Section 2.5 (Table 2) confirm that `human_only` is the dominant disagreement region: the model misses 111 tasks the human cohort solves, more than twice the size of the reverse case. Figure 1 visualizes the per-task split between the two agents.

When the two agents err, they err with different shapes. Table 3 breaks down VARC's failure modes: two thirds of VARC's wrong answers are `close_miss` — shape-correct grids that miss

	both_correct	human_only	varc_only	both_wrong
# tasks	173	111	48	68

Table 2: Quadrant counts. `human_only` is the largest disagreement region.

only a fraction of cells — with the remaining third split between `wrong_size` and `wrong_content`. Figure 2 shows the joint distribution alongside the human side. Conditional on being wrong, mean cell-level accuracy is 0.876 for VARC and 0.739 for humans (Figure 3). The model thus tends to land near the right structure with mismatched cells, while human errors more often replace content wholesale.

VARC error type (when wrong)	**Share**
close_miss	0.670
wrong_size	0.179
wrong_content	0.151

Table 3: VARC top-1 error type distribution conditional on the task being wrong. Most failures preserve grid shape and replace only a minority of cells.

3.2 VARC's view distribution tracks model confidence, not task difficulty

We first examine the VARC side alone. Conditioning on the four quadrants (Table 4), we find a sharp split: when VARC gets the task right (whether or not humans do), its 510 views agree on a small handful of grids; when VARC gets the task wrong, the 510 views scatter across hundreds of distinct outputs.

	top-1 share	top-1/top-2 gap	vote entropy	# unique grids
both_correct	0.644	0.581	1.818	81.4
varc_only	0.618	0.557	1.997	91.6
human_only	0.125	0.071	4.724	263.7
both_wrong	0.110	0.065	4.914	280.9

Table 4: VARC 510-view distribution metrics by quadrant. The top-1 share collapses by an order of magnitude on the two model-wrong rows, but the two model-wrong rows are nearly indistinguishable from each other.

Crucially, the VARC view distribution does *not* separate `human_only` from `both_wrong`: the entropy values (4.72 vs 4.91) and unique-grid counts (264 vs 281) are within sampling noise. Whatever signal the 510-view spread carries, it reflects whether the model itself has localized an answer, not whether the task is in principle solvable. Figure 4 visualizes this: human accuracy varies widely within the high-entropy regime, with `human_only` (red) and `both_wrong` (gray) overlapping heavily.

3.3 Human and model converge in different ways

We now bring in the human-side answer distribution. For each task we compute the same metrics over participants' final answer grids. The results, summarized in Table 5 and Figure 5, are striking.

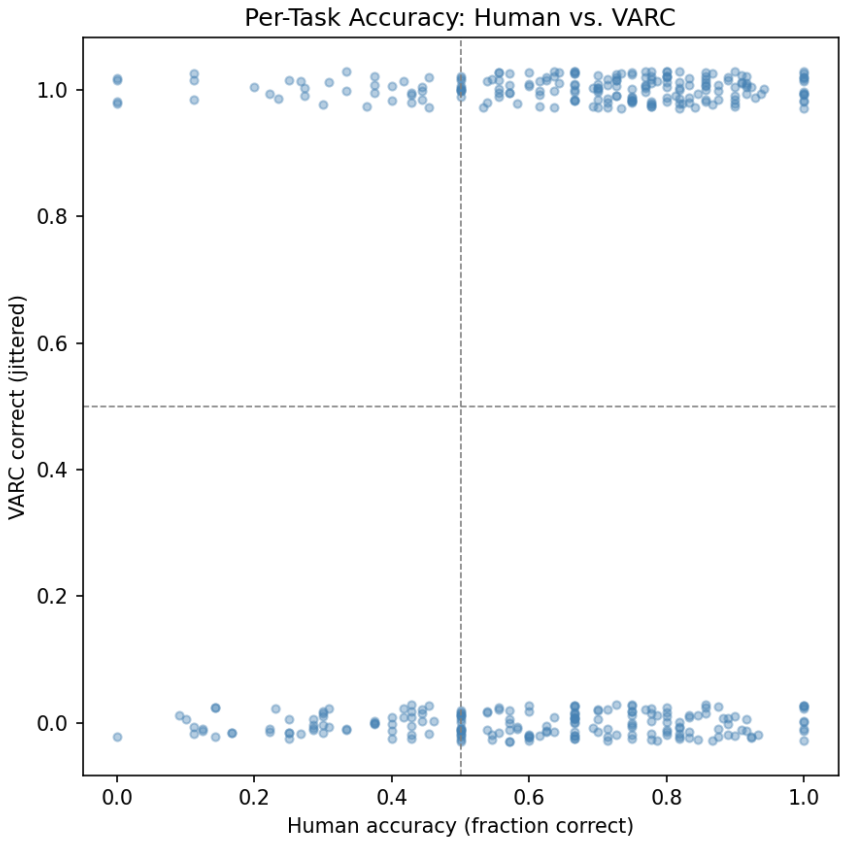

Figure 1: Per-task accuracy. Each point is one of the 400 evaluation tasks; the y-axis is binary VARC correctness with vertical jitter, the x-axis is human accuracy. Points cluster on the four corners defined by the quadrants of Section 2.5.

Two patterns are worth highlighting. First, on `human_only` tasks, the unique-grid count differs by nearly two orders of magnitude (≈ 3.6 vs ≈ 264). Humans don't merely outperform; they do so by entertaining a different size of hypothesis set. Second, `both_wrong` tasks show humans *also* disperse (entropy 1.77, ≈ 7 distinct grids) — but still an order of magnitude less than VARC's ≈ 281 grids. Even on tasks no one can solve, the human candidate set is bounded; the model's is not.

The Pearson correlation between per-task VARC entropy and human entropy is 0.21. This is non-trivial but small: tasks that confuse humans and tasks that confuse the model are largely different tasks. The two agents have different difficulty profiles, not a shared one with one agent uniformly weaker.

4 Discussion and Conclusion

4.1 Failure-to-induce

The strongest reading of our results is that the two agents are not failing at the same operation. Human errors — both on `human_only` (when participants disagree on a partially correct answer) and on `both_wrong` (when no participant succeeds) — sit in a small hypothesis space: at most a

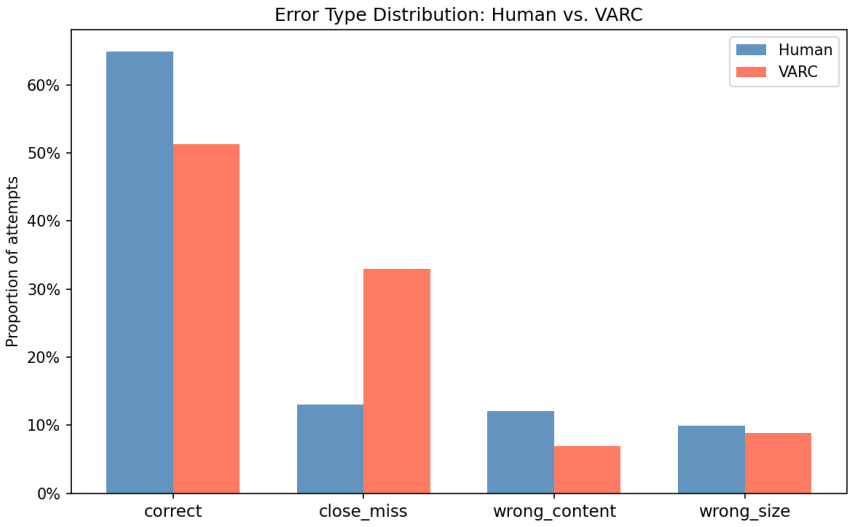

Figure 2: Error type distribution for VARC and humans across all attempts. The shapes of the two distributions differ: VARC clusters in `close_miss`, humans spread more evenly across error categories.

	VARC top-1	VARC entropy	Human top-1	Human entropy
both_correct	0.644	1.818	0.774	0.716
human_only	0.125	4.724	0.741	0.800
varc_only	0.618	1.997	0.357	1.752
both_wrong	0.110	4.914	0.355	1.767

Table 5: Top-1 share and Shannon entropy of VARC and human answer distributions, per quadrant. The `human_only` row is the defining contrast: humans concentrate (top-1 share 0.74, entropy 0.80) while VARC scatters (top-1 share 0.13, entropy 4.72).

handful of distinct grids, typically reflecting plausible but conflicting transformation rules. VARC's failures, by contrast, are nearly uniform over hundreds of candidate grids per task. The model is not picking the wrong rule from a shortlist; there is no shortlist.

We label this signature *failure-to-induce*. The model's multi-view procedure samples augmented presentations of the same input, and a system that had localized a transformation rule should be roughly view-invariant on it; the rule should produce a stable answer regardless of the augmentation. The fact that the 510 views disagree across hundreds of distinct outputs on the model's failure tasks suggests no such rule has been localized.

4.2 Why VARC's own confidence isn't enough

A natural alternative diagnostic would be to use the model's own view-level disagreement as a confidence signal — low entropy meaning "model knows," high entropy meaning "model doesn't know." Section 3.2 shows this signal is real but limited: it cleanly separates VARC-correct from VARC-wrong, but it cannot separate `human_only` from `both_wrong`. The model is equally confused on tasks that have a human-accessible solution and on tasks that don't. Self-reported confidence does not, on its own, identify the diagnostically interesting failures; it requires the human-side

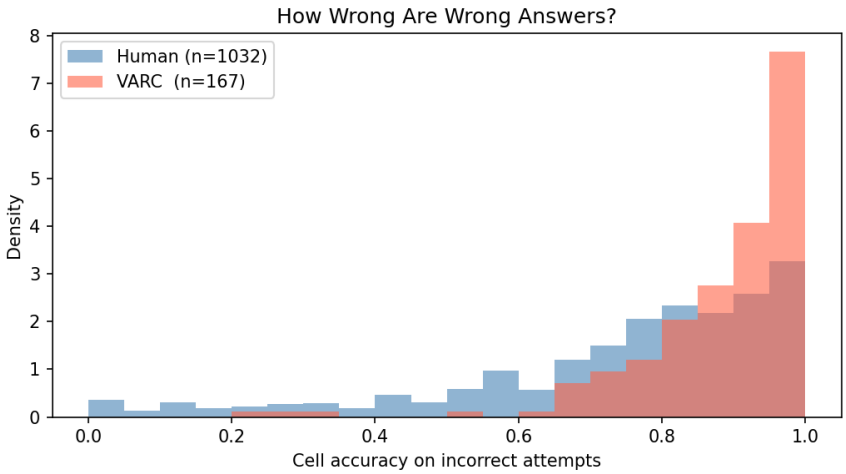

Figure 3: Cell-level accuracy distribution conditional on the task being wrong. VARC's mass concentrates near 1.0: most wrong predictions miss only a few cells. Human wrong attempts spread across the range, with many low-accuracy submissions reflecting wholesale content replacement.

distribution as a baseline.

4.3 Limitations

Our intervention is passive: we exploit VARC's existing 510-view sampling rather than designing controlled input edits (color permutation, rotation, etc.). This means we can characterize the *shape* of model failure but cannot causally identify the missing invariance. Second, we use only the plain ViT `attempt_0` configuration, not the ensemble setup described in the original VARC paper; the ensemble may be sharper. Third, the human distribution per task is built from ≈ 10 participants, which limits the resolution of the human entropy estimate.

4.4 Future work

A controlled active-intervention follow-up is the natural sequel: applying targeted augmentations (color permutation, rotation, mirror, scale) on the `human_only` subset and measuring how the view-level prediction set responds. Given that Section 3.2 shows VARC is already maximally diffuse on these tasks, the question is whether *any* structured intervention sharpens the distribution; if not, that itself is evidence that the model's failure is upstream of any specific invariance. A second direction is to use the entropy gap between humans and the model as a difficulty signal for constructing harder ARC evaluation sets.

4.5 Conclusion

On the ARC tasks that ordinary humans solve and the VARC ViT model fails, the two agents are not failing in the same way. Humans converge: their final answers cluster on a handful of grids, with one clearly modal. The model wanders: its 510 view-level predictions spread across hundreds of distinct grids, with no candidate exceeding 13% of the vote. The cognitive difference is not a matter of sharper precision; it is a matter of *whether a hypothesis set has been formed at all.*

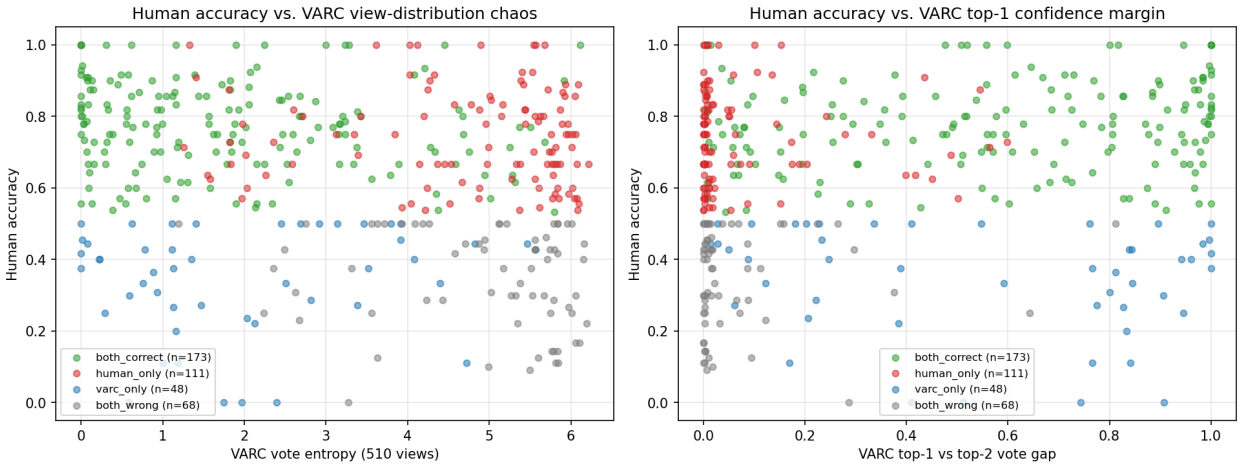

Figure 4: Human accuracy versus VARC view-distribution chaos. VARC's own confidence (entropy on the left, top-1/top-2 margin on the right) cleanly separates VARC-correct (green/blue) from VARC-wrong (red/gray), but does not separate `human_only` (red) from `both_wrong` (gray) within the model-wrong band.

References

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[2] K. Hu, A. Cy, L. Qiu, X. D. Ding, R. Wang, Y. E. Zhu, J. Andreas, and K. He, ARC Is a Vision Problem!, arXiv:2511.14761, 2025. `https://arxiv.org/abs/2511.14761`.

[3] S. LeGris, W. Vong, B. M. Lake, and T. M. Gureckis, H-ARC, OSF, 2025. `https://doi.org/10.17605/OSF.IO/BH8YQ`.

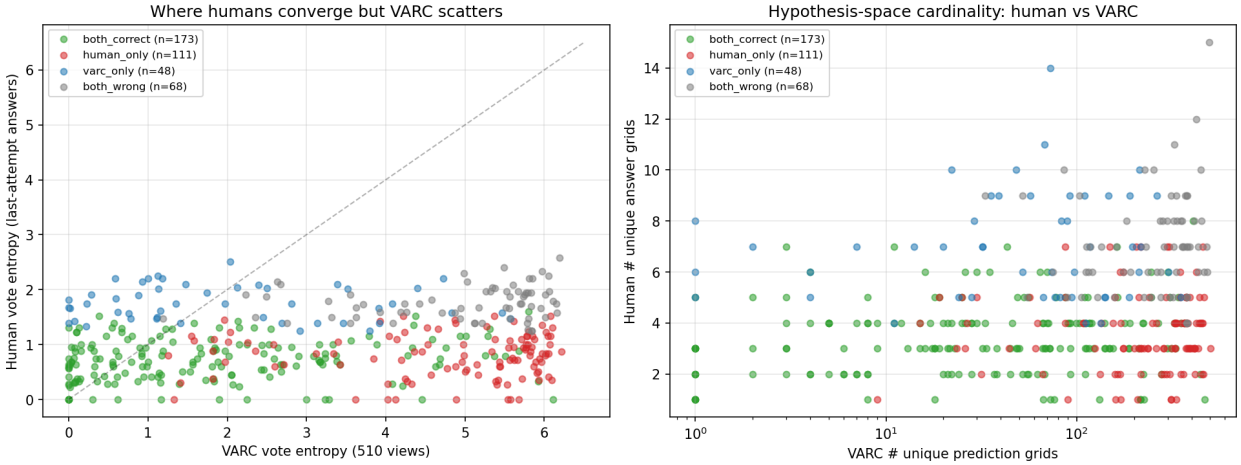

Figure 5: Direct comparison of VARC and human answer distributions per task. *Left:* entropy of VARC's 510-view distribution against entropy of the human last-attempt distribution. The dashed line is $y = x$. Red points (`human_only`) cluster far below the diagonal: humans converge while the model scatters. *Right:* number of distinct grids on each side; note the log scale on the VARC axis. `human_only` tasks place humans in the 2–5 grid regime and VARC in the 100–500 grid regime.